

Pandas Dataframes

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What is pandas? Python dictionaries, Dataframes? Basic operations,



Can talk about hash tables, dicts, hashable objects
e exemplificar com o novo built-in frozendict type in Python 3.15

<https://medium.com/data-science-collective/python-dictionaries-unraveled-the-hash-table-you-use-every-day-but-never-see-c5befda3beb0>

Advanced example

- § “Groups” using pandas dataframe
 - § <https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.groupby.html>
- § Example to avoid data leakage in biomedical datasets such as the example in next slides

show me a short example of groupby to group all personal info of a patient ID in a biomedical dataset
<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.groupby.html>

Yes. If your goal is simply to **keep all rows belonging to each patient together**, you can use `groupby` without aggregating anything.

For example, suppose your biomedical dataset has:

</> Python



▶ Run

```
import pandas as pd

df = pd.DataFrame({
    "patient_id": [101, 102, 101, 103, 102],
    "age":        [57, 63, 57, 45, 63],
    "sex":        ["M", "F", "M", "F", "F"],
    "glucose":    [110, 95, 125, 88, 102],
    "heart_rate": [72, 80, 75, 68, 78]
})
```



1. Group all information by patient

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```
groups = df.groupby("patient_id")
```

Now, for example:

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```
patient_101 = groups.get_group(101)
```

gives all records for patient 101:

	patient_id	age	sex	glucose	heart_rate
0	101	57	M	110	72
2	101	57	M	125	75



`groupby()` creates a `DataFrameGroupBy` object that lets you operate on each patient separately.



Pandas





2. If you want a dictionary: patient → all data

This is sometimes convenient:

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```
patients = {  
    patient_id: group  
    for patient_id, group in df.groupby("patient_id")  
}
```

Then:

</> Python



▶ Run

```
patients[101]
```

returns all records for patient 101.

3. Particularly useful for ML

For a biomedical dataset where you want **patient-level separation**, you might do:

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```
for patient_id, patient_data in df.groupby("patient_id"):
    print(patient_id, patient_data.shape)
```

This is useful, for example, when doing **leave-one-patient-out cross-validation**, because you can treat everything belonging to one `patient_id` as a single group rather than accidentally putting records from the same patient into both training and test sets.